LoDE - The logic of Entity Bases (EBs)

(HP2T)

LoDE – The Logic of Entity Bases

- **Intuition**
- Definition
- Domain
- Language
- Interpretation function
- Theories
- Entailment
- Reasoning problems
- Key notions

LoDE – Why a logic of Entity Bases?

- The **Logic of Entities (LoDE)** is the logic formalizing the reasoning implemented inside **Entity Bases (EBs)**, the most expressive formalization of knowledge in reasoning systems
- EBs are world models which result from the composition of CGs, ETGs, and EGs.
- LoDE extends the LoE reasoning by integrating it with reasoning about the underlying language (as from LoC) and about the underlying knowledge (as from LoD)
- With respect to LoE, LoDE presents two main improvements
  - It allows to define the meaning of words, as encoded in LoC definitions
  - It allows to describe the data and object properties of etypes an dtypes, as encoded in LoD descriptions, and to define etypes via LoD definitions.

LoDE – Highlights

- LoDE is a world logic, with both a graph linguistic / analogic representation.
- LoDE is the most expressive world logic which allows for a LoE-like KG representation of the information it encodes.
- It is conceptually similar in spirit to a earlier logic obtained from the union of the LoE Assertion Box (ABOX), what we call the EG, and the TBox terminologies of Description Logics (DL). The move is from DBs to KGs.

LoDE - Components

- The language used to name LoC concepts (LoC concept definitions)
- The language used to name LoD defined etypes (LoD etype definitions)
- The language used to describe complex etypes (LoD etype descriptions)
- The names of the entities, and their etypes which are the case in the world (LoE entity names)
- The knowledge (data and object properties) which describe how entities correlate in the world (LoE etype descriptions)

Entity Bases

The original use of the term **knowledge base** was to describe one of the two sub-systems of an **expert system** (a **knowledge-based system**).

As from [*] **knowledge-based system** consists of

- a **knowledge-Base** (**KB**) representing ***facts***** about the world and
- ways of **reasoning** about those ***facts*** to deduce new facts or highlight inconsistencies.

We call an **Entity Base** (**EB**) a KB where assertions are stored as an **Entity Graph**.

** Hayes-Roth, Frederick; Donald Waterman; Douglas Lenat (1983). <u>Building Expert Systems</u>. Addison-Wesley.*

*** That is, in our terminology, **assertions** denoting **facts***

EB – Intuition

$$EB = < L,T,A>$$

where:

- ***L*** (for ***L***exicon) is a terminology which formalizes the meaning of words into a lexicon that we use to describe the world. We formalize ***L*** in LoC.
- ***T*** (for ***T***eleontology) is a terminology which formalizes the background (commonsense, scientific, …) knowledge of the world. We formalize ***T*** in LoD.
- ***A*** (for ***A***ssertions) is an assertional theory which models our knowledge of the world as we perceive it, as we are told about it, as we infer about i. We formalize ***A*** in LoE.

CGs, ETGs and EGs provide, respectively, the means for represenying about ***L*** **,** ***T*** and ***A***.

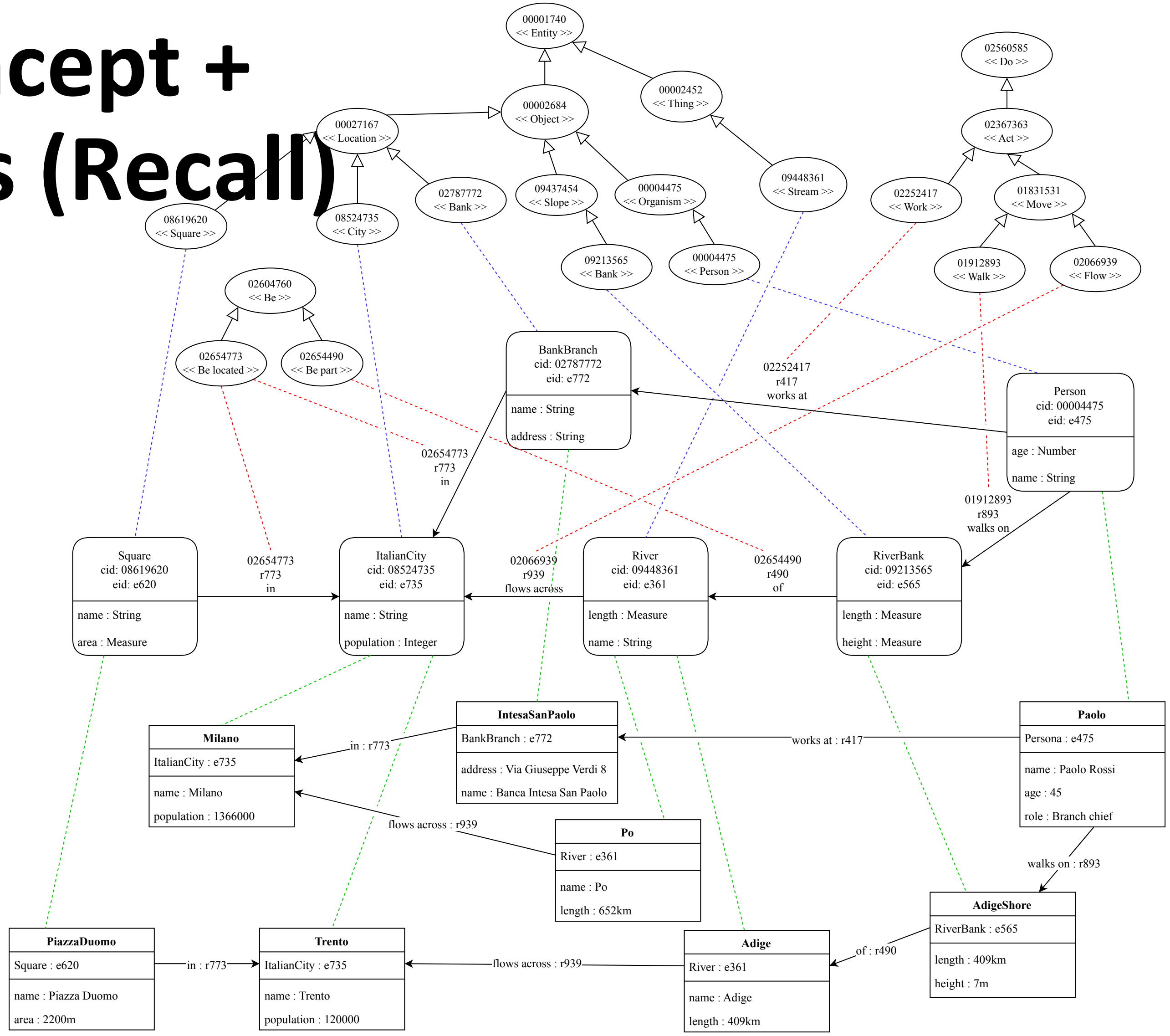
LoC, LoD and LoE provide, respectively, the means for reasoning about ***L*** **,** ***T*** and ***A***.

LoDE provide the means for reasoning about *A* based on the knowledge encoded in *T*, using the language used in *L*.

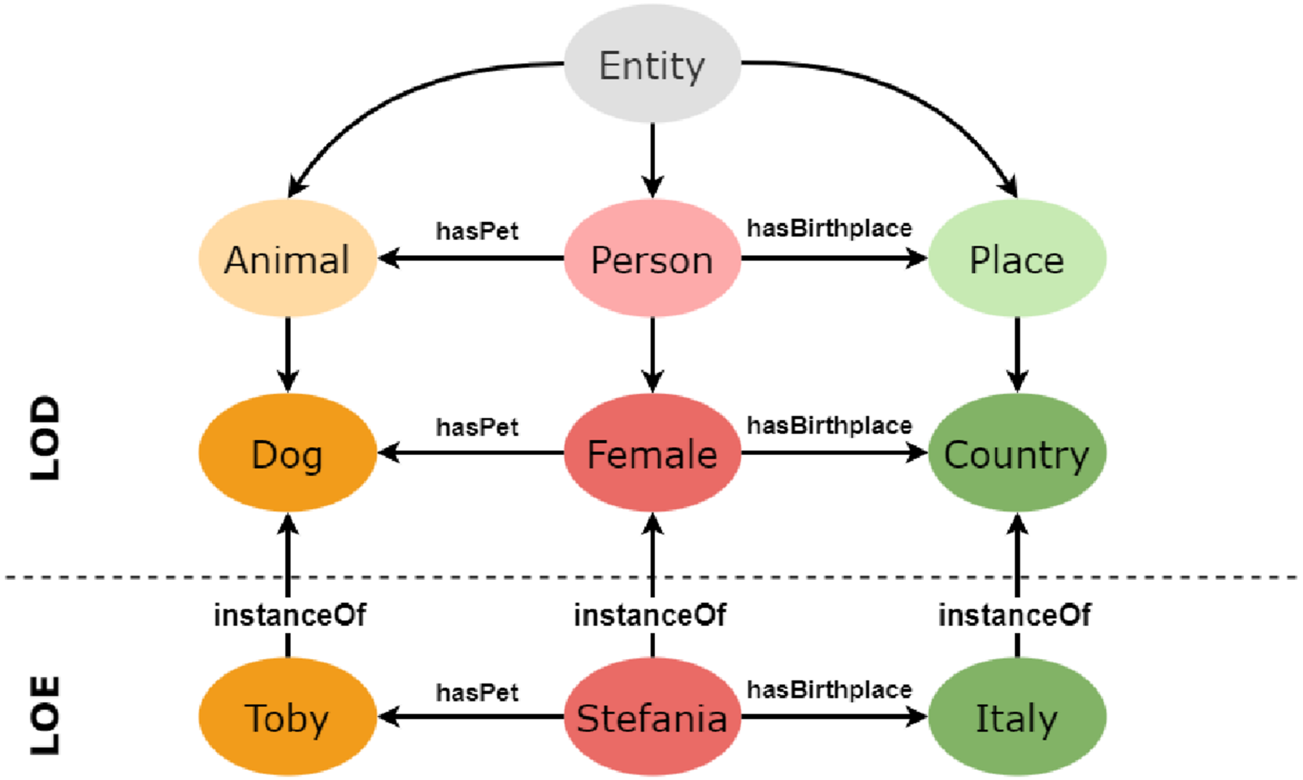
World models as concept + etype + entity graphs (Recall)

entity graphs
which are complete
with respect to their
reference
etype graphs
and **concept graphs**

The information represented in these models cannot be computed from etype + entity graphs



An example of EB Components



A <u>populated</u> knowledge teleontology

- *Which LoD definitions?*
- *Which LoD descriptions?*
- *Which LoE assertions?*

Could we have an EG encoding the following assertions?

- Person (Lucia)
- HasBirthplace (person#1, Italy)
- Person(Stefania)
- HasPet(Stefania, Toby)
- HasPet(Stefania, Dog#2)

Observation. In an EB the goal is to provide answers about the properties of entities in the LOE EG. The LOD knowledge component is used «only» to make assertions informed by the LOD implicit knowledge.

Well-formedness conditions

An EG, to be well-formed must satisfy the following conditions:

- Each node is associated to one and only ONE entity or value;
- Each node is associated to one and ONLY one etype/ dtype;
- Each link is associated with one and only one data or object property;
- Etypes and dtypes must be grounded (via MG/LG links) in concepts
- Data and object properties must have the correct etypes or datatypes (strong typing);
- Data and object properties must be concepts ;
- No links are allowed starting from data value nodes.

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LoDE – Definition

We formally define the logic LoDE as follows

$$\mathbf{LoDE} = \langle \mathbf{EG_{LoDE}}, \vDash_{\mathbf{LoDE}} \rangle$$

with

$$\mathbf{EG_{LoDE}} = \langle \mathbf{L_{LoDE}}, \mathbf{D_{LoDE}}, \mathbf{I_{LoDE}} \rangle$$

Below, any time no confusion arises, we drop the subscripts.

LoDE – Definition (continued)

Intuition (LoDE Logic). LoDE = $\langle EG_{LoDE}, \vDash_{LoDE} \rangle$ is defined via a suitable composition of

$$LoE = \langle EG_{LoE}, \vDash_{LoE} \rangle,$$
$$LoD = \langle EG_{LoD}, \vDash_{LoD} \rangle,$$
$$LoC = \langle EG_{LoC}, \vDash_{LoC} \rangle,$$

where, $EG_{LoDE} = \langle L_{LoDE}, D_{LoDE}, I_{LoDE} \rangle$ is constructed by suitably composing

$$EG_{LoE} = \langle L_{LoE}, D_{LoE}, I_{LoE} \rangle,$$
$$ETG_{LoD} = \langle L_{LoD}, D_{LoD}, I_{LoD} \rangle,$$
$$ETG_{LoC} = \langle L_{LoC}, D_{LoC}, I_{LoC} \rangle,$$

and the same with $\vDash_{LoDE}$.

LoDE – Definition (continued)

Intuition (LoDE Logic). The key intuitions, and follow-up assumptions, as detailed in the following of this section, underlying how LoDE is composed from LoE, LoD, LoC are:

- The universe of interpretation, that is, the entities considered, are the same for all the component logics, that is: $U_{LoDE} = U_{LoE} = U_{LoD} = U_{LoC}$.
- The concepts in L_{LoC} are types (etypes or dtypes) or (data and object) properties in $L_{LoD.}$ We call this type of etypes, dtypes and properties, **primitive**. All dtypes and properties are primitive (they are predefined).
- The types (etypes and dtypes) and properties in L_{LoE} are L_{LoD} types and properties.
- The LoD types (and correspondingly LoC concepts) are selected in order to provide a proper contextualization and domain-aware specification of the LoE assertions.

LoDE – Definition (continued)

Observation (LoDE Logic). The assumptions mentioned above guarantee that:

- all the statements about the real world made in LoE and describing the intended model must be well grounded into the LoD types and properties
- all the LoD types and properties must be grounded the into the LoC concepts and
- LoD and LoC are used to provide the proper domain and context for a suitable representation of the world being modeled, as depicted by the intended model.

Observation (LoDE Logic). Concerning the third point, as it will be described later, LoDE lets the user free to decide everything, including which concepts to consider, which etypes to map to concepts, the level of detail of the representation and so on.

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Domain (the same as LoE)

Definition (LoDE Domain)

$$D = < U, \{C\}, \{R\} >$$

where:

- $U = \{u\}$ is called the **universe (of interpretation)** of D.
- $\{u\}$ is a set of **units** $u_1, \ldots, u_n$, for some n
- $\{C\}$ is a set of **classes** $C_1, \ldots, C_m$ of units, for some m, with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \ldots, R_p$ between units, for some p, with $R_i \subseteq U \times U$

Observation (LoE, LoD, LoC domains). The LoDE domain is the same as LoE and it is representationally richer than LoD (which does not represent units) and LoC (which represents only {C}).

Universe of interpretation (the same as LoE)

Definition (LoDE Universe of interpretation)

$$U = E \cup V$$

where:

- $U = \{u\}$ is the **universe of interpretation**;
- $E = \{e\} \subseteq \{u\}$ is the **entity universe**;
- $V = \{v\} \subseteq \{u\}$ is the **value universe**;
- $\{e\}$ and $\{v\}$ are disjoint.

Observation (LoE, LoD, LoC Universes). The universe of interpretation is inherited by LoE. Note that LoD indirectly mentions the units in U via the two quantifiers while LoC does not refer, implicitly or explicitly, to the units in U.

Classes (the same as LoE, LoD, LoC)

Definition (LoDE Class)

$$\{C\} = E_T \cup D_T$$

where:

- $E_T = \{E_i\}$ is a **set of etypes** E_i, with $E_i = \{e\} \subseteq E$
- $D_T = \{D_i\}$ is a **set of dtypes** D_i, with $D_i = \{v\} \subseteq V$

Observation (etype). For any etype E_i, $E_i \subseteq E$.

Observation (dtype). For any dtype D_i, $D_i \subseteq V$

Observation (LoE LoD, LoC classes). The LoDE classes are inherited from LoD which, in turn, inherits primitive types from LoC. Loc defines what exists in the current formalization of the world

Binary relations (the same as LoE, LoD)

Definition (LoDE Binary relation)

$$\{R\} = O_R \cup A_R$$

where:

- $O_R = \{O_i\}$ is a **set of object properties** O_i, with $O_i \subseteq E_k \times E_j$
- $A_R = \{A_i\}$ is a **set of attributes** A_i, with $A_i \subseteq E_k \times D_j$

Observation (LoE, LoD, LoC binary relations). All binary relations are defined as concepts (denoting pairs of types) in LoC, defined as binary relations in LoD, and used in LoE

An example of domain of LoDE (continued)

E = {#1, #2, #3, #4, #5, #6, #7, #8, #9, 1,85m, 1,78m, 15/05/1990, http://www.paolo.it}

E_T = {Entity, Person}

D_T = {Data, Real, Boolean, String, Date, Length}

{R} = {hF, hD, hH, hB, hL, hU}

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Language (the same as LoE, LoD, LoC)

Definition (Assertional language)

$$L_a = \langle A_a, FR_a \rangle = \{a\}$$

where:

- L_a is an **assertional language**
- A_a is an **alphabet** of assertions
- FR_a is a **set of formation rules**
- $\{a\}$ is the set of assertions obtained by the exhaustive application of FR_a to A_a (the transitive closure $FR_a(A_a)$ of FR_a applied to A_a).

Observation (LoD, LoC alphabet). The LoDE set of formation rules FR_a is the union of the sets of formation rules of LoE, LoD and LoC. They are recursively applied bottom up starting from the assertiosn generated by the LoE formation rules.

Alphabet (the same as LoE)

Definition (Alphabet A)*

$$\mathrm{A}_a = < E, \{T\}, \{P\} >$$

where:

- $E = \{e\} \cup \{v\}$ is a set of (names of) **entities** e and of values v;
- $\{T\} = \{E_i\} \cup \{D_i\}$ is a set of unary predicates standing for **etypes** and **dtypes**;
- $\{P\} = \{O_i\} \cup \{A_i\}$ is a set of binary **properties**, where O_i is an **object property**, also called a **role**, and A_i is an **attribute**.

Observation (LoE, LoD, LoC alphabet). The LoDE alphabet is the same as LoE but it is a superset of LoD (which does not contain entities) and LoC (which contains only {C}). LoD and LoC have the possibility of extending their alphabet.

*The elements of the alphabet are written in *italic* to distinsguish them from percepts

LoDE formation rules – BNF

$$
\begin{aligned}
<a_{\text{LoDE}}> &::= <a_{\text{LoE}}> \;|\\
&\; <a_{\text{LoD}}> \;|\\
&\; <a_{\text{LoC}}> \;|\\
<a_{\text{LoD}}> &::= <a_{\text{LoDef}}> \;|\\
&\; <a_{\text{LoDes}}>
\end{aligned}
$$

Observation (LoDE formation rules). The set of the LoDE Formation rules are the union of sets of formation rules of LoE, LoD and LoC.

Formation Rules – BNF (almost the same as LoE)

<a_{LoE}> ::= <etype>(<nameEntity>) |
<dtype>(<value>) |
<objProp>(<nameEntity>, <nameEntity>) |
<dataProp>(<nameEntity>, <value>)

<etype> ::= E_1 | …| E_n | <defEtype>

<dtype> ::= D_1 | …| D_n

<objProp> ::= O_1 | …| O_n

<dataProp> ::= A_1 | …| A_n

<nameEntity> ::= e_1 | …| e_n

<value> ::= v_1 | …| v_n

where

- <defEtype> is a LoD etype which is defined or primitive (a Concept) *i= 1, …n*
- E_i, D_i, O_i, O_i, are LoC concepts, that is D_i, O_i, $O_i \in A_C = \{c\}$ *i= 1, …n*
- $e_i, v_i \in A_a$ are a LoE entity and a LoE value, respectively, *i= 1, …n*

Definitions and descriptions – BNF (the same as LoD)

$$
\begin{aligned}
< a_{\mathrm{LoDef}} > ::= \; & \langle \mathrm{defEtype} \rangle \sqsubseteq < p_C > | \\
& \langle \mathrm{defEtype} \rangle \equiv < p_C > | \\
& \langle \mathrm{defEtype} \rangle \perp < p_C > | \\
& \langle \mathrm{defEtype} \rangle \underline{\perp} < p_C > |
\end{aligned}
$$

$$
\begin{aligned}
< a_{\mathrm{LoDes}} > ::= \; & < p_C > \sqsubseteq < p_C > | \\
& < p_C > \equiv < p_C > | \\
& < p_C > \perp < p_C > | \\
& < p_C > \underline{\perp} < p_C > |
\end{aligned}
$$

with $< p_C >$ being LoD composite etype percepts , that is $< p_C > \in L_C = \{p_C\}$

Composite etype formation rules – BNF (a subset of LoD)

$$< p_C > \quad ::= < p_C > \sqcap < p_C >$$
$$< p_C > \quad ::= < p_T >$$

Observation (LoDE LoD-like formation rules). In LoDE it is not possible to represent disjunction. Negative information can only be represented using disjointness in assertions. That is, it is not possible to have nested disjointness information.

Observation (LoDE LoD-Like formation rules). The restriction above still allows to encode the knowledge encoded in world models (see later), given the extended language for definitions and descriptions (see previous slide).

Formation rules – BNF (Recall)

Notation (BNF). $< p_C >$ is a nonterminal symbol and it stands for a p_C percept. $<p_T>$ is an L_C terminal symbol and stands for an L_T percept. See above how the BNF of L_T expands to a LoD terminal symbol.

Observation (BNF). This BNF does allow the iterative application of the formation rules. It allows to generate percepts of any depth.

Observation (Conjunctive composite etypes). The intuition is that the conjunctive etype $< p_C > \sqcap < p_C >$ has the properties of both the component etypes. Semantically it means thatcomposite its elements are elements of both the component etypes.

Observation (Disjunctive composite etypes). The intuition is that the disjunctive etype $< p_C > \sqcup < p_C >$ has the properties of at least one of the component etypes. Semantically it means that its elements are elements of one or both the component etype.

Observation (Complement composite etypes). The intuition is that the complement etype $\neg < p_C >$ has the opposite properties of at least one of the component etypes. Semantically it means that its elements are all the elements of the universe of interpretation which are not elements of $< p_C >$.

Observation (Composite etypes and world models). Composite etypes cannot be represented in the world models seen the past. This is the case in particular for disjunctive and complement etypes.

Etype formation rules – BNF (the same as LoD)

< p_{T} > ::= <etype> | <dtype> |

< etype > ::= ∃<objProp> .<etype> |
∃<dataProp>.<dtype> |
∀<objProp> .<etype> |
∀<dataProp> .<dtype>

<etype> ::= E_1 | ...| E_n

<dtype> ::= D_1 | ...| D_n

<objProp> ::= O_1 | ...| O_n

<dataProp> ::= A_1 | ...| A_n

Formation Rules – BNF (the same as LoC)

$$
\begin{aligned}
< a_{\mathrm{LoC}} > \quad ::= \quad & c_1 \sqsubseteq c_2 \;| \\
& c_1 \equiv c_2 \;| \\
& c_1 \perp c_2 \;| \\
& c_1 \underline{\perp} c_2
\end{aligned}
$$

with $c_1, \ldots, c_n$ being LoC concepts, that is $c_1, \ldots, c_n \in A_C = \{c\}$

LoDE Assertions – as from the BNF

Observation. LoDE assertions only have one of sixteen possible forms:

- Four LoE assertions
- Four plus four LoD assertions
- Four LoC assertions

LoE assertions – as from the BNF (Recall)

Observation. LoE assertions only have one of four possible forms:

- $E_T(e)$, meaning that the entity e is of etype E_T
- $D_T(v)$, meaning that the value v is of dtype D_T
- $O(e_1, e_2)$, meaning that the object property O holds between e_1 and e_2
- $A(e, v)$, meaning that the data property A of entity e has value v

LoD assertions – as from the BNF (Recall)

Observation. LoD assertions have one of four possible forms:

- $< p_C > \sqsubseteq < p_C >$, meaning that the first composite percept etype is subsumed by the second composite percept etype
- $< p_C > \equiv < p_C >$, meaning that the first composite percept etype is equivalent to the second composite percept etype
- $< E > \sqsubseteq < p_C >$, meaning that the first etype is defined as being subsumed by the composite percept etype
- $< E > \equiv < p_C >$, meaning that the first etype is defined as being equivalent to the composite percept etype

Four more LoD assertions – as from the BNF

Observation. LoD, as used in LoDE, allows for the following four more assertions:

- $< p_C > \perp < p_C >$, meaning that the first composite percept etype is disjoint with the second composite percept etype
- $< p_C > \underline{\perp} < p_C >$, meaning that the first composite percept etype is strongly disjoint with the second composite percept etype
- $< E > \perp < p_C >$, meaning that the etype is disjoint with the composite percept etype
- $< E > > \underline{\perp} < p_C >$, meaning that the etype is strongly disjoint with the composite percept etype

LoC assertions – as from the BNF (Recall)

Observation. LoC allows for the following four assertions:

- $c_1 \sqsubseteq c_2$, meaning that the first concept is subsumed by the second concept
- $c_1 \equiv c_2$, meaning that the first concept is equivalent to the second concept
- $c_1 \perp c_2$, meaning that the first concept is disjoint with the second concept
- $c_1 \underline{\perp} c_2$, meaning that the first concept is strongly disjoint with the second concept

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LoDE – Interpretation function

Definition (Interpretation function I_{LoDE}):

$$I_{LoDE} = \langle\, I_{LoC}, I_{LoD}, I_{LoE} \,\rangle$$

Observation: I_{LoD} and I_{LoC} are first applied to compute the extension the of defined etypes and concepts, which are then used to compute the semantics of LoE assertions (via I_{LoE}).

Interpretation of EG assertions (same as LoE)

Definition (Interpretation function, alphabet). The **LoE interpretation function I** is defined as

$$I_{LoE} : L \rightarrow D$$

where I is defined as

$$I_{LoE} = < I_U, I_T, I_P >$$

where:

- $I_U = <I_e, I_v>$, the universe interpretation function, is the pair of the entity and the value interpretaton function;
- $I_T = <I_E, I_D>$, the Type interpretation function, is the pair of the etype and the dtypes interpretaton function;
- $I_P = <I_O, I_A>$, the Property interpretation function, is the pair of the object property and the attribute interpretaton function

with:

$$I_e : \{e\} \rightarrow \{\mathrm{e}\} \qquad I_v : \{v\} \rightarrow \{\mathrm{v}\}$$
$$I_E : \{E_i\} \rightarrow \{\mathrm{E_i}\} \qquad I_D : \{D_i\} \rightarrow \{\mathrm{D_i}\}$$
$$I_O : \{O_i\} \rightarrow \{\mathrm{O_i}\} \qquad I_A : \{A_i\} \rightarrow \{\mathrm{A_i}\}$$

Interpretation of EG assertions (same as LoE)

Definition (LoE interpretation function, assertion). The interpretation of assertions into facts is defined as follows:

$$I_{LoE}(E_i\ (e)) = I_E(E_i)\ (I_e(e)) = e \in E_i$$
$$I_{LoE}(D_i\ (v)) = I_D(D_i)\ (I_v(v)) = v \in D_i$$
$$I_{LoE}(O_i\ (e_1,e_2)) = I_O(O_i)\ (I_e(e_1), I_e(e_2)) = \langle e_1, e_2 \rangle \in O_i$$
$$I_{LoE}(A_i\ (e,v)) = I_A(A_i)\ (I_e(e), I_v(v)) = \langle e, v \rangle \in A_i$$

Observation. Notice how the interpretation of assertions first applies the interpretation function (i.e., I_{LoE}) which then, **compositionally**, applies the proper component specific interpretation function mimicking, step by step how syntax composes (e.g., in the first assertion, I_E and I_e).

Notation. The words of the alphabet are in "*italic*", percepts are in "roman"

Interpretation of ETG descriptions (same as LoD)

$I_{LoD}(p_1 \sqsubseteq p_2)$ iff $I_c(p_1) \subseteq I_c(p_2)$

$I_{LoD}(p_1 \equiv p_2)$ iff $I_c(p_1) = I_c(p_2)$

iff $I_c(p_1) \subseteq I_c(p_2)$ and $I_c(p_2) \subseteq I_c(p_1)$

$I_{LoD}(p_1 \perp p_2)$ iff $I_c(p_1) \cap I_c(p_2) \subseteq \emptyset$

$I_{LoD}(p_1 \underline{\perp} p_2)$ iff $I_c(p_1) \cap I_c(p_2) \subseteq \emptyset$ and $I(p_1) \cup I(p_2) \subseteq U$

Where I_c is the intepretation function for L_c, the language of LoD composite etype percepts

Interpretation of ETG definitions (continued)

$I_{LoD}(E \sqsubseteq p_2)$ iff $I_c(E) \subseteq I_c(p_2)$

$I_{LoD}(E \equiv p_2)$ iff $I_c(E) = I_c(p_2)$

iff $I_c(E) \subseteq I_c(p_2)$ and $I_c(p_2) \subseteq I_c(E)$

$I_{LoD}(E \perp p_2)$ iff $I_c(E) \cap I_c(p_2) \subseteq \emptyset$

$I_{LoD}(E \underline{\perp} p_2)$ iff $I_c(E) \cap I_c(p_2) \subseteq \emptyset$ and $I(E) \cup I(p_2) \subseteq U$

Where I_c is the intepretation function for L_c, the language of LoD composite etype percepts

Interpretation of composite etype percepts (a subset of LoD)

$$\mathrm{I_C}(p_1 \sqcap p_2) = \mathrm{I_C}(p_1) \cap \mathrm{I_C}(p_2)$$

$$\mathrm{I_C}(p_\mathrm{T}) = \mathrm{I_T}(p_\mathrm{T})$$

$$\mathrm{I_T}(p_\mathrm{T}) = \mathrm{p_T}$$

where:

- $\mathrm{I_C}$ is the interpretation function of $\mathrm{L_C}$, the LoD language for composite types
- $\mathrm{I_T}$ is the intepretation function for $\mathrm{L_T}$, the LoD language of primitive etypes.
- p_1, p_2 are composite etype percepts
- p_T (in *italic*) is (the name of an) etype percept denoting the domain percept $\mathrm{p_T}$ (not in italic)

Interpretation of etype percepts (same as LoD)

$I_T(\top)$ = U, with U the universe of interpretation

$I_T(\perp)$ = Ø, with Ø the empty set

$I_T(E_i) = E_i$

$I_T(D_i) = D_i$

$I_T(\exists P.T) = \{d \in U \mid$ there is an $e \in U$ with $(d, e) \in I_T(P)$ and $e \in I_T(T)\}$

$I_T(\forall P.T) = \{d \in U \mid$ for all $e \in U$ if $(d, e) \in I(P)$ then $e \in I_T(T)\}$

where I_T is the interpretation function of L_T, the LoD language of primitive etypes and dtypes.

Interpretation of CG definitions (same as LoC)

$I_{LoC}(c_1 \sqsubseteq c_2)$ iff $I_c(c_1) \subseteq I_c(c_2)$

$I_{LoC}(c_1 \equiv c_2)$ iff $I_c(c_1) = I_c(c_2)$

iff $I_c(c_1) \subseteq I_c(c_2)$ and $I_c(c_2) \subseteq I_c(c_1)$

$I_{LoC}(c_1 \perp c_2)$ iff $I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$

$I_{LoC}(c_1 \underline{\underline{\perp}} c_2)$ iff $I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$ and $I(c_1) \cup I(c_2) \subseteq U$

with $c_1, c_2 \in A_C = \{c\}$

LoDE facts – as from the BNF

Observation. LoDE facts only have one of sixteen possible forms:

- Four LoE facts
- Four plus four LoD facts
- Four LoC facts

LoE Facts (Recall)

Observation. LoE facts only have one of four possible forms:

- $e \in E_i$, that is, the entity e has etype E
- $v \in D_i$, that is, the value v has dtype D
- $\langle e_1, e_2 \rangle \in O_i$, that is, the object property O_i holds between the two entities e_1, e_2
- $\langle e, v \rangle \in A_i$, that is, the attribute A of entity e has value v

LoD Facts (Recall)

Observation. LoD facts only have one of four possible forms:

- $p_1 \subseteq p_2$, meaning that the first composite percept is subsumed by the second composite percept
- $p_1 = p_2$, meaning that the first composite percept is the same as the second composite percept
- $E \subseteq p_2$, meaning that the etype is defined as being subsumed by the composite percept
- $E = p_2$, meaning that the etype is defined as being the same as the composite percept

Four more LoD Facts

Observation. LoD, as used in LoDE, allows for the following four more facts:

- $p_1 \cap p_2 \subseteq \emptyset$, meaning that the first composite percept etype is disjoint with the second composite percept etype
- $p_1 \cap p_2 \subseteq \emptyset$ and $p_1 \cup p_2 = U$, meaning that the first composite percept etype is disjoint with the second composite percept etype and their union is the universe of interpretation
- $E \cap p_2 \subseteq \emptyset$, meaning that the etype is disjoint with the composite percept etype
- $E \cap p_2 \subseteq \emptyset$ and $p_1 \cup p_2 = U$, meaning that the etype is disjoint with the composite percept etype and their union is the universe of interpretation

LoC Facts (Recall)

Observation. LOC allows for the following four facts:

- $c_1 \subseteq c_2$, the extension of a concept is a subset of the extension of another concept.
- $c_1 = c_2$, the extension of a concept is the same as that of another concept.
- $c_1 \cap c_2 \subseteq \emptyset$, the extension of a concept is disjoint with that of another concept.
- $c_1 \cap c_2 \subseteq \emptyset$ and $c_1 \cup c_2 = U$, the extension of a concept is disjoint with that of another concept and their union is the universe of interpretation.

An example of domain of LoDE (continued)

We have the following example including an EG, with its own reference ETG , the ETG with it own reference CG

E = {#1, #2, #3, #4, #5, #6, #7, #8, #9, 1,85m, 1,78m, 15/05/1990, http://www.paolo.it}

E_T = {Entity, Person}

D_T = {Data, Real, Boolean, String, Date, Length}

{R} = {hF, hD, hH, hB, hL, hU}

from which we construct the following facts in the domain

D = {#1 ∈ Entity, 1,85∈ Real, 1,85 ∈ Data, hF(#1, #1), …,
P ⊆ entity, Real ⊆ data, hF(P, P), hD(P, D), hH(P, Real), …}

with, e.g., hF(P, P) standing for hF ⊆ P x P

LoDE – The Logic of Entity Bases

- Intuition
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- **Theories**
- Entailment
- Reasoning problems
- Key notions

World models – basic and composed types (Recall)

Three basic types of world models (**base** world models)

- Concept graphs (CG)
- Etype (entity type) graphs (ETG)
- Entity graphs (EG)

Four types of compositions of world models (**composed** world models)

- Concept + etype graphs
- Concept + entity graphs
- Etype + entity graphs
- Concept + etype + entity graphs

KG Theories

Observation (Seven types of world models). As from above there seven types of world models, that is

- (Base) Concept graphs (CG)
- (Base) Etype (entity type) graphs (ETG) /* two subtypes: Teleontology, Teleology */
- (Base) Entity graphs (EG)
- (Composed) Concept + etype graphs
- (Composed) Concept + entity graphs
- (Composed) Etype + entity graphs
- (Composed) Concept + etype + entity graphs

Definition (KG theory of a world model). Given a world model a KG theory is an assertional theory, that is, a set of assertions, written in the proper world logic (LoE, LoD, LoC) language, which has the same set of models.

Observation (Seven types of KG theories). There are seven types of kG theories, one per world model

From world models to KG Theories

Proposition (World model to KG theory). For any world model it is possible to construct a KG theory for that world model.

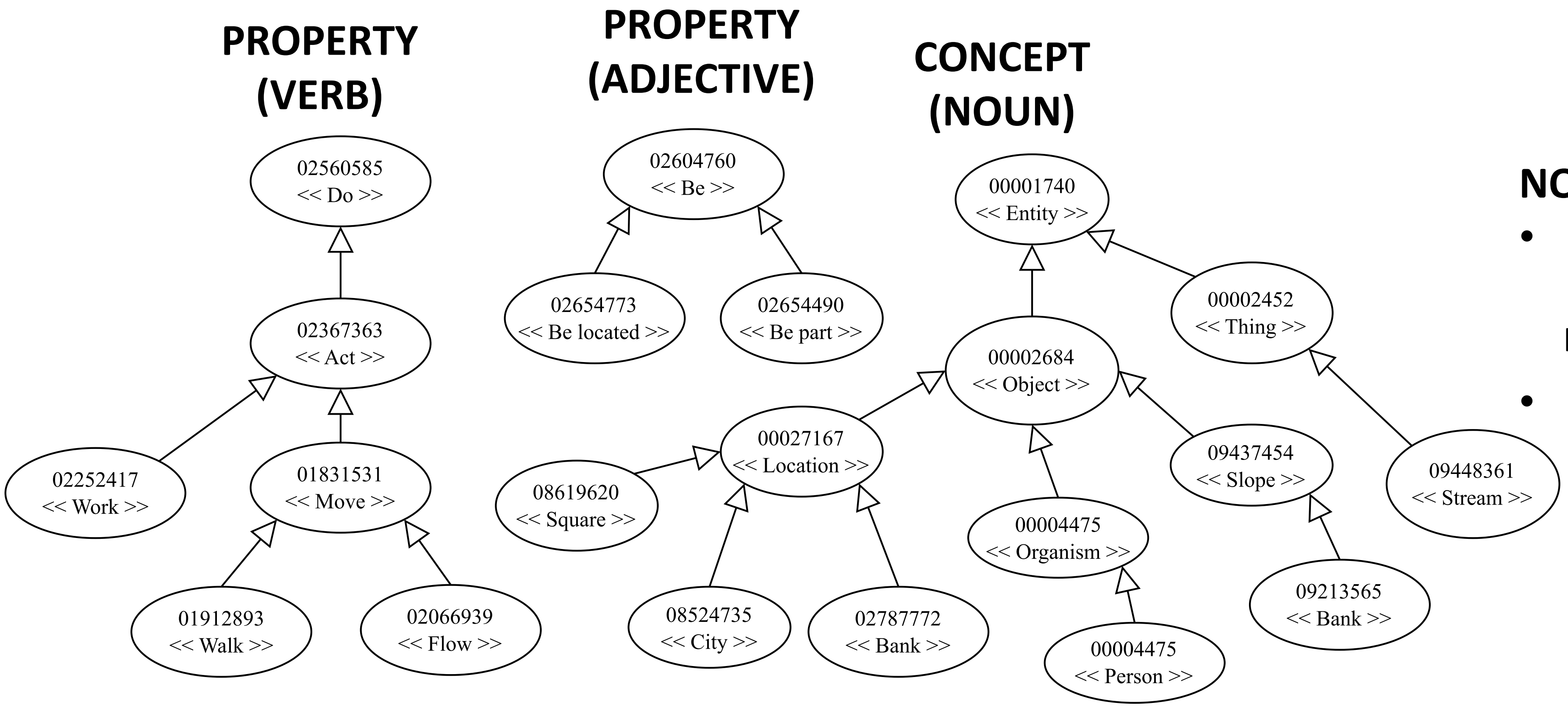
Proposition evidence (Base world model to KG theory). One by one, for each link in the base world model, write a assertion in the proper world logic language and then take the union of the resulting assertions.

Proposition evidence (Composed world model to KG theory). Build the KG theories from the component base world models. Then take the union of the resulting KG theories.

Observation (Composed world model to KG theory). To be able to represent KG theories in LoDE you need to have the proper (LoDE) language (see above)

Observation (LoDE Language). Disjunction («⊔») and negation («¬ ») are not needed to build KG theories which encode logically the information encoded in world models.

Concept graphs – example (Recall)



NOTE:

- All siblings are pairwise disjoint, e.g.,
  I(organism) ∩ I(location) = ∅
- All children are subsets of the parents, e.g.,
  I(city) ⊆ I(location)

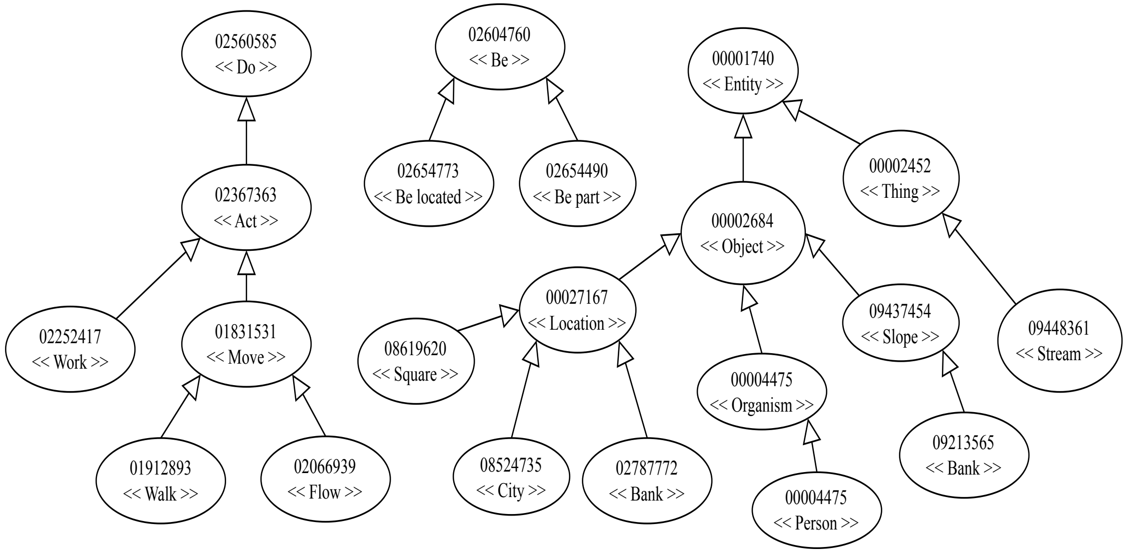
PROPERTY
(ADJECTIVE)

A LoC KG theory – example

PROPERTY
(VERB)

PROPERTY
(ADJECTIVE)

CONCEPT
(NOUN)



CONCEPT
(NOUN)

$Object \sqsubseteq Entity$

$Thing \sqsubseteq Entity$

$Object \perp Thing$

$Stream \sqsubseteq Thing$

$Slope \sqsubseteq Object$

$Organism \sqsubseteq Object$

$Location \sqsubseteq Object$

$Slope \perp Organism$

$Slope \perp Location$

$Organism \perp Location$

$Bank1 \sqsubseteq Slope$

$Person \sqsubseteq Organism$

$Square \sqsubseteq Location$

$City \sqsubseteq Location$

$Bank2 \sqsubseteq Location$

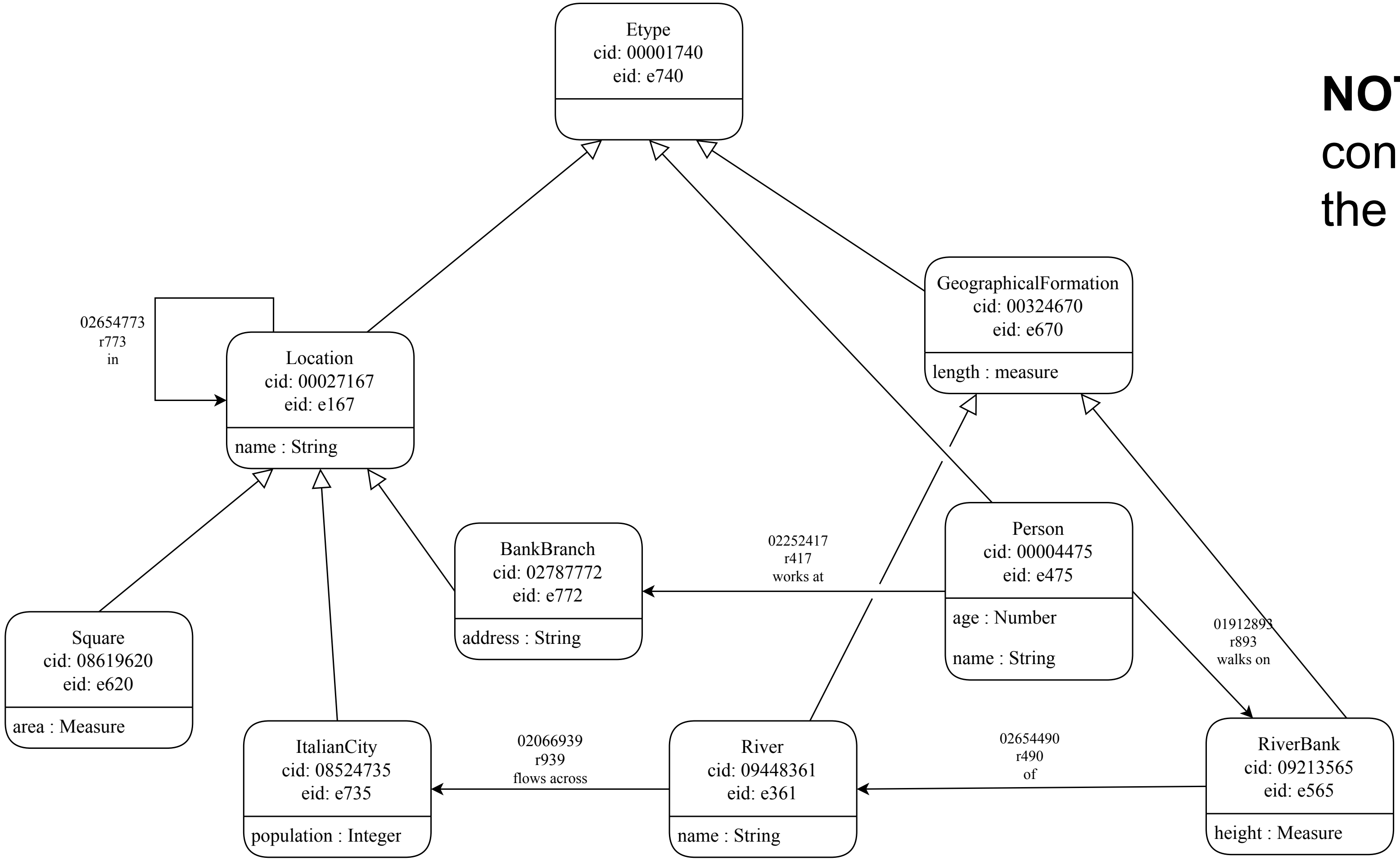
$Square \perp City$

$Square \perp Bank2$

$Bank2 \perp City$

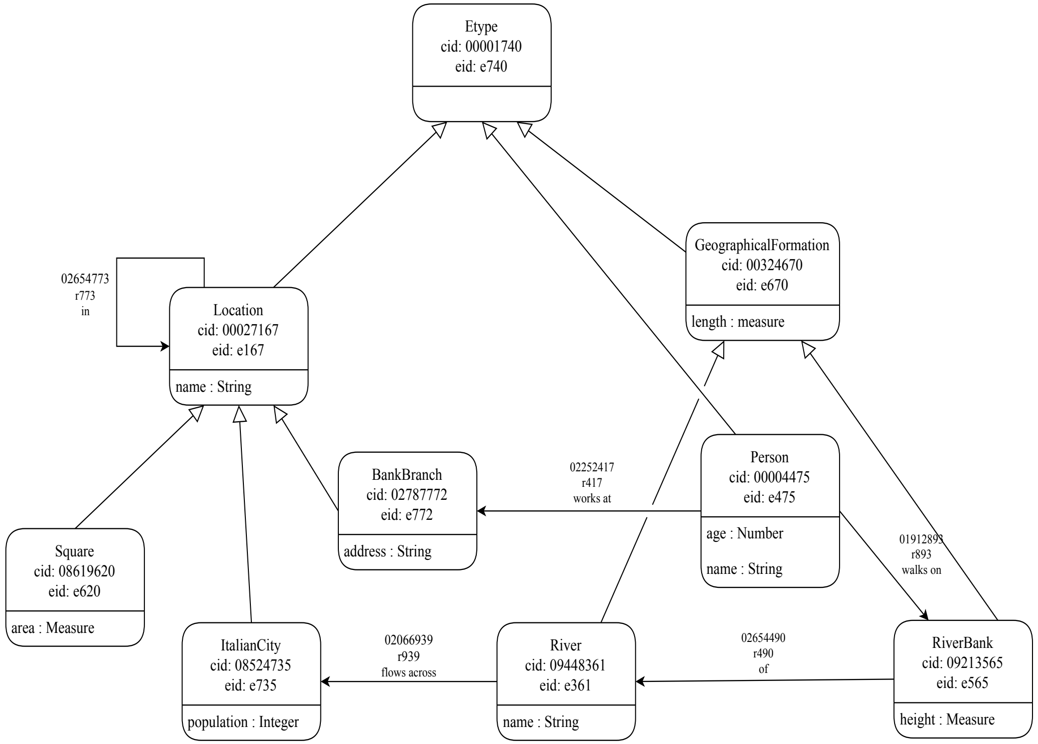
PROPERTY
(VERB)

Knowledge teleontology (KTLO) – example (Recall)



NOTE: Cid's of types connect to the Cid of the reference concept

A LoD KG theory – example



$$GeographicalFormation \sqsubseteq Entity \sqcap \exists length.Measure$$

$$Location \sqsubseteq Entity \sqcap \exists name.String \sqcap \forall in.Location$$

$$Person \sqsubseteq Entity \sqcap \exists age.Number \sqcap \exists name.String \sqcap \forall worksAt.BankBranch \sqcap \forall walksOn.RiverBank$$

$$BankBranch \sqsubseteq Location \sqcap \exists address.String$$

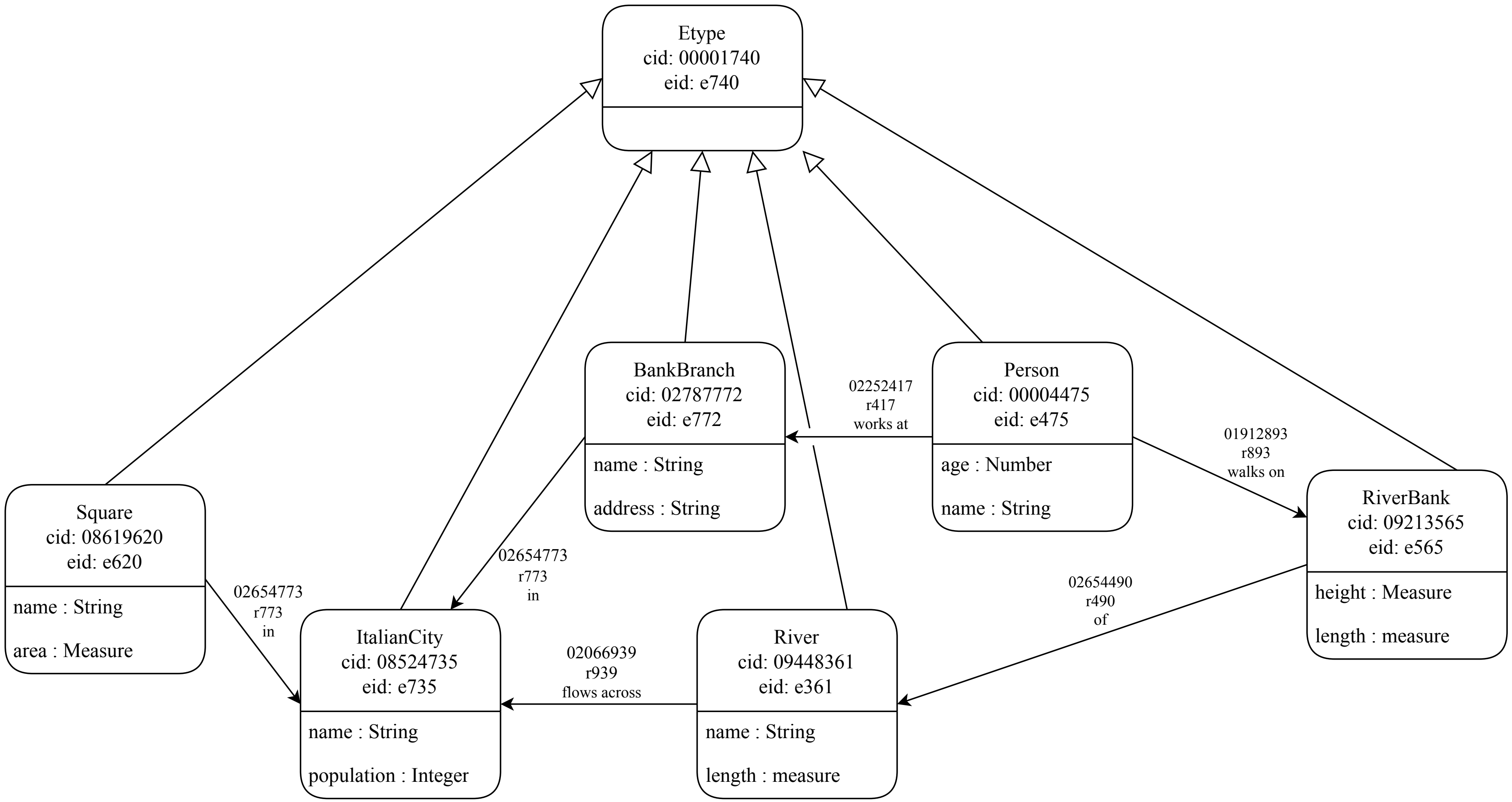
$$Square \sqsubseteq Location \sqcap \exists area.Measure$$

$$ItalianCity \sqsubseteq Location \sqcap \exists population.Integer$$

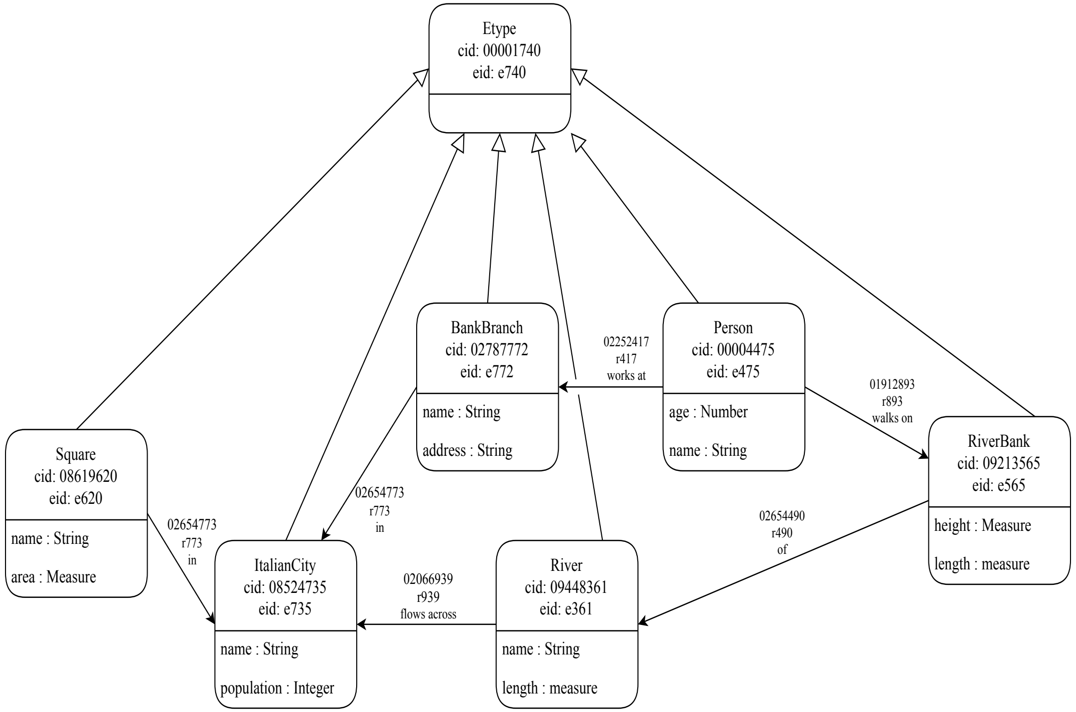
$$River \sqsubseteq GeographicalFormation \sqcap \exists name.String \sqcap \forall flowAcross.ItalianCity$$

$$RiverBank \sqsubseteq GeographicalFormation \sqcap \exists height.Measure \sqcap \forall of.River$$

Teleology (TLO) – example (Recall)



A LoD KG theory – example (Recall)



$$Person \sqsubseteq Entity \sqcap \exists age.Number \sqcap \exists name.String \sqcap \forall worksAt.BankBranch \sqcap \forall walksOn.RiverBank$$

$$BankBranch \sqsubseteq Entity \sqcap \exists address.String \sqcap \exists name.String \sqcap \forall in.ItalianCity$$

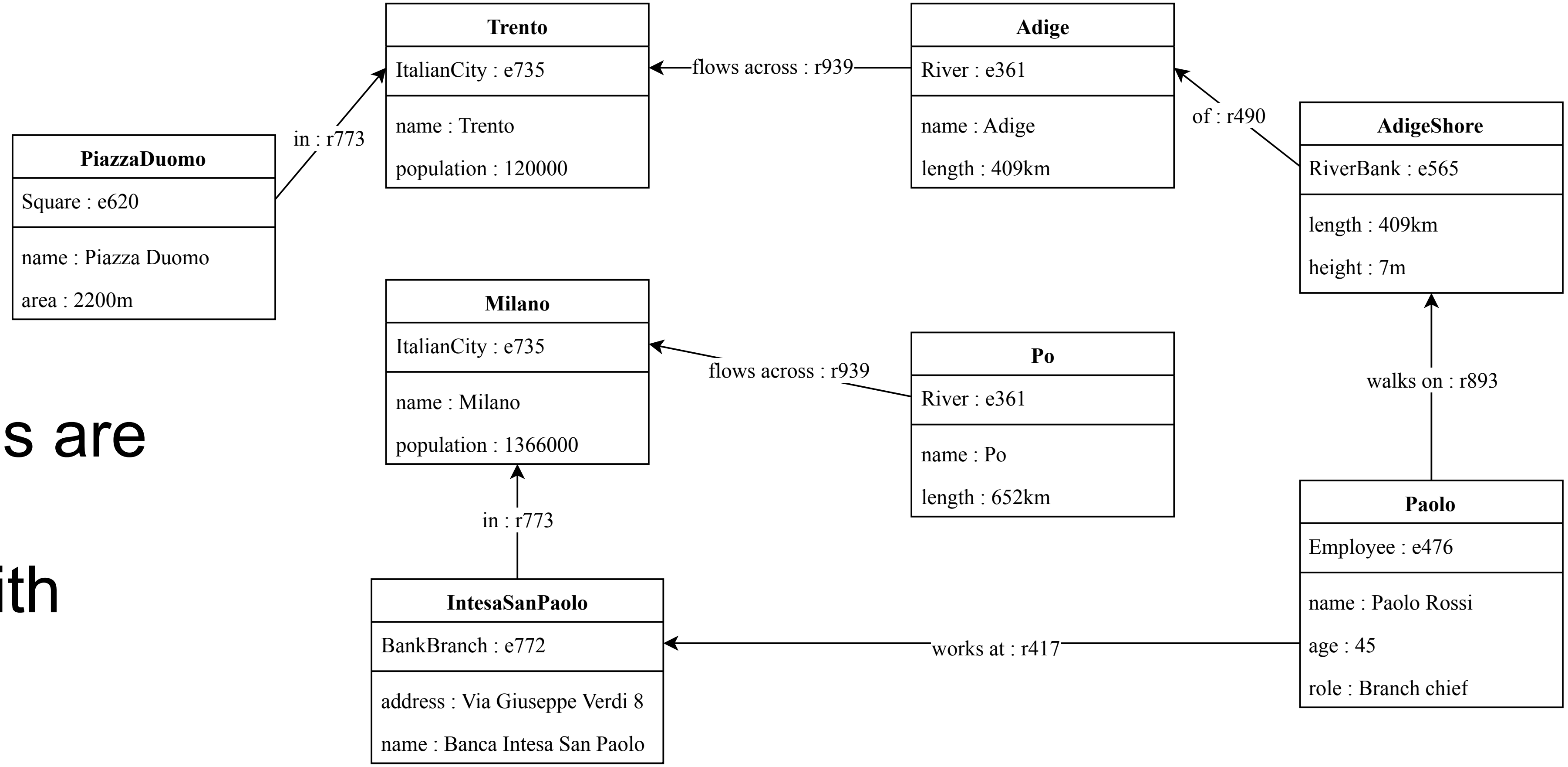
$$Square \sqsubseteq Entity \sqcap \exists area.Measure \sqcap \exists name.String \sqcap \forall in.ItalianCity$$

$$ItalianCity \sqsubseteq Entity \sqcap \exists population.Integer \sqcap \forall in.Location$$

$$River \sqsubseteq Etype \sqcap \exists name.String \sqcap \forall flowAcross.ItalianCity \sqcap \exists length.Measure$$

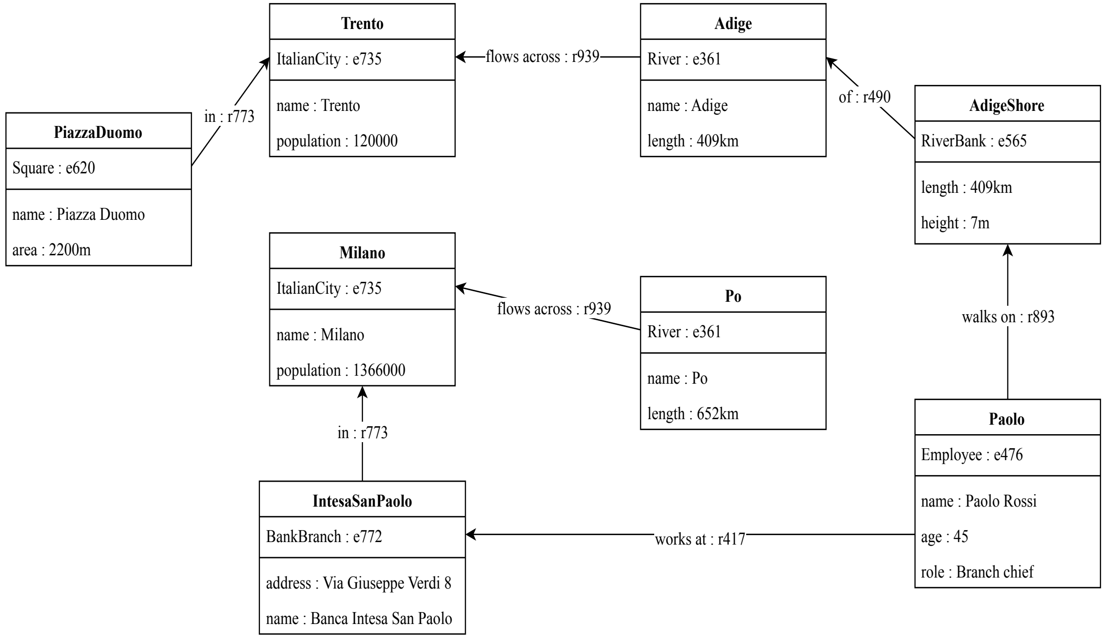
$$RiverBank \sqsubseteq Entity \sqcap \exists length.Measure \sqcap \exists height.Measure \sqcap \forall of.River$$

Entity graph (Recall)



Entity Graphs are schemas populated with entities

A LoE KG Theory - example



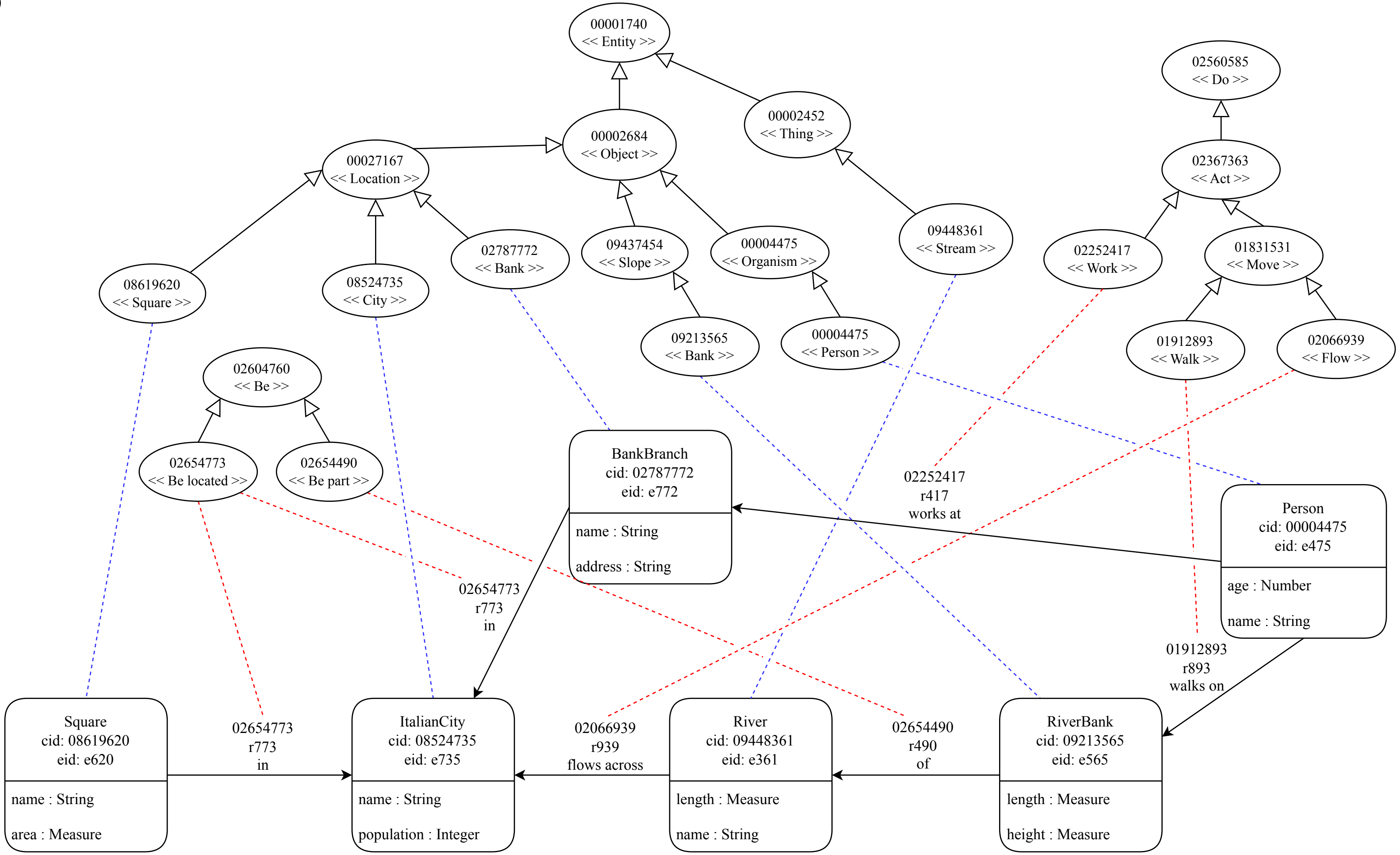
$Square(PiazzaDuomo)$
$ItalianCity(Trento)$
$ItalianCity(Milano)$
$River(Adige)$
$River(Po)$
$RiverBank(AdigeShore)$
$Person(Paolo)$
$BankBranch(IntesaSanPaolo)$
$String(\text{“Piazza Duomo”})$
$String(\text{“Trento”})$
$String(\text{“Adige”})$
$String(\text{“Paolo Rossi”})$
$String(\text{“Milano”})$
$String(\text{“Po”})$
$String(\text{“Via Giuseppe Verdi 8”})$
$String(\text{“Banca Intesa San Paolo”})$
$Measure(2200m)$
$Measure(409km)$
$Measure(7m)$
$Measure(652km)$
$Number(120000)$
$Number(1366000)$
$Number(45)$

$name(PiazzaDuomo, \text{"Piazza Duomo"})$
$name(Trento, \text{"Trento"})$
$name(Adige, \text{"Adige"})$
$name(Paolo, \text{"Paolo Rossi"})$
$name(Po, \text{"Po"})$
$name(Milano, \text{"Milano"})$
$name(IntesaSanPaolo, \text{"Banca Intesa San Paolo"})$
$area(PiazzaDuomo, 2200m)$
$population(Trento, 120000)$
$population(Milano, 1366000)$
$length(Adige, 409km)$
$length(AdigeShore, 409km)$
$length(Po, 652km)$
$height(AdigeShore, 7m)$
$age(Paolo, 45)$
$address(IntesaSanPaolo, \text{“Via Giuseppe Verdi 8”})$
$in(PiazzaDuomo, Trento)$
$in(IntesaSanPaolo, Milano)$
$flowsAcross(Adige, Trento)$
$flowsAcross(Po, Milano)$
$of(AdigeShore, Adige)$
$walksOn(Paolo, AdigeShore)$
$worksAt(Paolo, IntesaSanPaolo)$

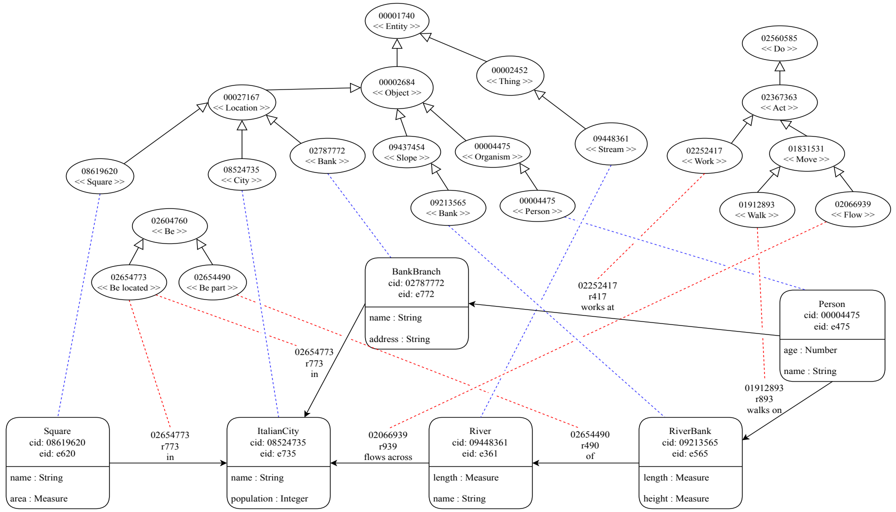
World models as concept + etype graphs (Recall)

Etype graphs which are complete with respect to their reference **concept graphs**

The information represented in these models cannot be computed from etype graphs



A LOCD KG Theory - example



$Object \sqsubseteq Entity$

$Thing \sqsubseteq Entity$

$Object \perp Thing$

$Stream \sqsubseteq Thing$

$Slope \sqsubseteq Object$

$Organism \sqsubseteq Object$

$Location \sqsubseteq Object$

$Slope \perp Organism$

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$Person \sqsubseteq Organism$

$Square \sqsubseteq Location$

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$Bank2 \sqsubseteq Location$

$Square \perp City$

$Square \perp Bank2$

$Bank2 \perp City$

$$Person \sqsubseteq Entity \sqcap \exists age.Number \sqcap \exists name.String \sqcap \forall worksAt.BankBranch \sqcap \forall walksOn.RiverBank$$

$$BankBranch \sqsubseteq Entity \sqcap \exists address.String \sqcap \exists name.String \sqcap \forall in.ItalianCity$$

$$Square \sqsubseteq Entity \sqcap \exists area.Measure \sqcap \exists name.String \sqcap \forall in.ItalianCity$$

$$ItalianCity \sqsubseteq Entity \sqcap \exists population.Integer \sqcap \forall in.Location$$

$$River \sqsubseteq Etype \sqcap \exists name.String \sqcap \forall flowAcross.ItalianCity \sqcap \exists length.Measure$$

$$RiverBank \sqsubseteq Entity \sqcap \exists length.Measure \sqcap \exists height.Measure \sqcap \forall of.River$$

LoDE – The Logic of Entity Bases

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World entailment (Recall)

Definition (World entailment) Let W = < L_a, D, I_a > be a world model. Let L_a be an assertional language. Then $|=_{La}$ is a **world entailment relation**

$$\models_{La} \subseteq D \times L_a$$

such that

$$M \models_{La} T_a$$

M={f} ⊆ D **is a model of** $T_a \subseteq L_a$. We say that M **entails** T_a. We write $M \models T_a$ instead of $M \models_{La} T_a$ when no confusion arises.

LoE World Entailment (Recall)

Definition (LoE world entailment). Let T_a be a LoE theory for a model M. Let $a \in L_a$ and $T_a \subseteq L_a$ be respectively an **assertion question** and a **theory question**. Then:

$M \vDash T_a$ if and only if $M \vDash a$, for all $a \in T_a$

$M \vDash \{a\}$ if and only if $M \vDash a$, written $M \vDash a$

LoD world entailment (Recall)

$M \models p_1 \sqsubseteq p_2$ iff $I(p_1 \sqsubseteq p_2)$ iff $I(p_1) \subseteq I(p_2)$

$M \models p_1 \equiv p_2$ iff $I(p_1 \equiv p_2)$ iff $I(p_1) = I(p_2)$

iff $I(p_1) \subseteq I(p_2)$ and $I(p_2) \subseteq I(p_1)$

with $p_1, p_2 \in L_{LoD}$.

LoD extended World Entailment

$$M \models p_1 \perp p_2 \quad \text{iff} \quad I_c(p_1) \cap I_c(p_2) \subseteq \emptyset$$

$$M \models p_1 \underline{\perp} p_2 \quad \text{iff} \quad I_c(p_1) \cap I_c(p_2) \subseteq \emptyset \text{ and } I(p_1) \cup I(p_2) \subseteq U$$

with $p_1, p_2 \in L_{LoD}$.

LoC World Entailment (Recall)

$M \models c_1 \sqsubseteq c_2$ iff $I_c(c_1) \subseteq I_c(c_2)$

$M \models c_1 \equiv c_2$ iff $I_c(c_1) = I_c(c_2)$

iff $I_c(c_1) \subseteq I_c(c_2)$ and $I_c(c_2) \subseteq I_c(c_1)$

$M \models c_1 \perp c_2$ iff $I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$

$M \models c_1 \underline{\perp} c_2$ iff $I_c(c_1) \cap I_c(c_2) \subseteq \emptyset$ and $I(c_1) \cup I(c_2) \subseteq U$

with $c_1, c_1 \in A_C = \{c\}$.

LoE, LoD, LoC, LoDE World entailment

Reminder (LoE world entailment). LoE world entailment checks for checks for **equality / disequality** between (the interpretation of) two assertions.

Reminder (LoD World entailment). LoD world entailment checks for **equality / disequality** and **similarity / dissimilarity** between (the interpretation of) two etypes. The intuition is that the LoD language allows for complex assertions (both definitions and descriptions). LoD entailment amounts to checking whether two percepts, costructed in different way actually name the same set or a smaller set or different sets.

Reminder (LoC World entailment). LoC world entailment checks for **equality** / **disequality** and **similarity / dissimilarity** between (the interpretation of) two concepts.

Proposition (LoDE World entailment). Depending on the input KG theory, which must encode a base world model, LoDE world entailment reduces to LoE, LoD, LoC world entailment.

Proposition (World entailment for composed world models). World entailment is not defined for KG theories which encode composed world models.

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Logical entailment (Recall)

Definition (Logical entailment). Let $L_W = \langle W, \vDash_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with T, $\underline{T} \subseteq L$ and $a \in L$. Then we write

$$T \vDash_{\{M\}} \underline{T} \qquad\qquad (T \vDash_{\{M\}} a)$$

and say that T **logically entails** $\underline{T}$ (T **logically entails** a) with respect to {M} if

$$\text{for all } M \in \{M\}, \text{ if } M \vDash T \text{ then } M \vDash \underline{T} \; (M \vDash a)$$

Observation (Model filtering via logical entailment). Logical entailment implements reasoning in the following way: "Whatever we derive from T must be true in the same sets of "relevant" models {M} as T. T acts as a filter which allows to discard all the irrelevant models.

Observation (Model filtering via selection of {M}). The choice of the models {M} of T is arbitrary. And it also amounts to model filtering. It could be the only model of T, all models of T, a model selected because it is the intended model of T, and so on. If $M \nvDash T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ({M}). When no confusion arises we write $\vDash$ instead of $\vDash_{\{M\}}$.

LoE – Reasoning problems (Recall)

Instance checking, Checking whether an assertion is entailed by a theory, i.e. checking whether

$$T_{LoDE} \models E(e)$$
$$T_{LoDE} \models P(e1, e2)$$

with E an etype or a dtype, P an object or a data property.

Observation (Instance retrieval). Instance retrieval is not considered as it is a multiple application of instance retrieval.

LoD – Reasoning problems (Recall)

Observation (LoD reasoning problems). The four LoD core reasoning problems are:

- $T \models C$ Satisfiability with respect to a TBox T
- $T \models C \sqsubseteq D$ Subsumption with respect to a TBox T
- $T \models C \equiv D$ Equivalence with respect to a TBox T
- $T \models C \perp D$ Disjointness with respect to a TBox T

where T can also be empty.

Observation (Empty TBox). All LoD reasoning problems are logical entailment problems. When the TBOx is empty, then LoD logical entailment reduces to LoD world entailment on the reference model(s).

LoC – Reasoning problems (Recall)

Observation (LoC reasoning problems). The four LoC core reasoning problems are:

- $T \vDash C$ Satisfiability with respect to a TBox T
- $T \vDash C \sqsubseteq D$ Subsumption with respect to a TBox T
- $T \vDash C \equiv D$ Equivalence with respect to a TBox T
- $T \vDash C \perp D$ Disjointness with respect to a TBox T

where T can also be empty.

Observation (LoC vs. LoD). LoC reasoning problems are the same as LoD's, but applied to concepts and not etypes with a language of much lower expressivity

LoDE – Reasoning problems

Definition (LoDE Reasoning problems for KG theories encoding base world models). Reduce logical entailment to the relevant world logic, viz., one of LoE, LoD, LoC.

Definition (Level of abstraction of a KG theory). We have the following

- A LoE theory is more abstract than a LoE theory
- A LoD theory is more abstract than a LoD, or LoE theory
- A LoC theory is more abstract than a LoC, or LoD, or LoE theory

Intuition (Level of abstraction of a KG theory). Language (LoC KG theories) is more abstract than knowledge (LoD KG theories) which, in turn, is more abstract than entities (LoE KG theories).

LoDE – Reasoning problems

Definition (LoDE Reasoning problems for KG theories). Let a reasoning problem be stated as

$$T \models_{\{M\}} C$$

where T is a more abstract KG theory than the KG theory C. Solve $M \models C$ in the Logic of C for all $M \in \{M\}$ such that $M \models T$, where $M \models T$ is solved in the logic of T.

Observation (KG theory reasoning problems). Based on the definition above, all reasoning problems associated with the seven KG theories encoding the seven base and composed world models can be solved in the LoDE world logic.

LoDE – Reasoning problems

Observation (LoDE Reasoning problems for KG theories). The above process can be iterated, that is, for any model M such that M ⊨ C in the Logic of C we can compute the entailment M ⊨ D, whee the logic of C is more abstract than the logic of D. This derives from the fact that a KG theory can be built incrementally from multiple models as long as the abstraction ordering is maintained.

Observation (LoDE Reasoning problems for KG theories – follow-up). The possibility of iterating the process of model entailment is quite useful when there are multiple world models and corresponding KG theories, e.g., one or more LoC KG theories, one or more LoD KG theories and one of more LoE KG theories

LoDE – Language and knowledge reasoning problems

Definition (LoDE language and knowledge reasoning problem). LoDE language and knowledge reasoning problems are reasoning problems $T \vDash_{\{M\}} C$ where both $M \vDash T$ and $M \vDash C$ are solved in LoD (LoC).

Observation (LoDE language and knowledge reasoning problem). LoDE language and knowledge reasoning problems apply to the KG theories of the following base and composed world models:

- LoC: Concept graphs (CG)
- LoD: Etype (entity type) graphs (ETG)
- LoCD (LoC+LoD): Concept + etype graphs (CETG)

LoDE – entity reasoning problems

Definition (LoDE Entity Reasoning problem). LoDE entity reasoning problems are reasoning problems $T \vDash_{\{M\}} C$ where $M \vDash T$ is solved in LoD (LoC) while $M \vDash C$ is solved in LoE.

Observation (LoDE entity reasoning problem). LoDE entity reasoning problems apply to the KG theories of the following base and composed world models

- LoE: Entity graphs (EG)
- LoCE (LoC + LoE):Concept + entity graphs (EG)
- LoDE (LoD + LoE): Etype + entity graphs (EETG)
- LoCDE(LoC+LoD+LoE): Concept + etype + entity graphs (CEETG)

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Key notions

- Knowledge Base
- Entity Base
- KG theories
- LoDE world entailment
- LoDE logical entailment
- Language reasoning problems
- Knowledge reasoning problems
- Entity reasoning problems

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Dipartimento di Ingegneria e Scienza dell'Informazione

Know dive

LoDE - The logic of Entity Bases (EBs)

(HP2T)

Sept 13,2023